Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/29/2015
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Risk of Intervertebral Disc Damage after Prolonged Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ahargens@ucsd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	858-534-7837
Organization Name:	University of California, San Diego		
PI Address 1:	Altman Clinical and Translational Research Institute		
PI Address 2:	9452 Medical Center Drive/0863		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
Start Date:	07/11/2013	**End Date:**	07/10/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Ploeger, Stephanne	**Contact Phone:**	
Contact Email:	stephanne.l.ploeger22@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lotz, Jeffrey (University of California, San Francisco) O'Neill, Conor (Self) Sayson, Jojo (Ola Grimsby Institute, San Diego) Chiang, Stephen (Methodist Hospital) Chang, Douglas (University Of California, San Diego) Allon, Moshe (Self) Parazynski, Scott (The Methodist Hospital Research Institute) Riascos-Castaneda, Roy (University of Texas Medical Branch)		
Grant/Contract No.:	NNX13AM89G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Editor's Note (5/2/2014): NOTE THIS IS A CONTINUATION OF FUNDING FOR NNX10AM18G WITH THE SAME TITLE AND PRINCIPAL INVESTIGATOR. Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (IRP Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on International Space Station-ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken 1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
Task Progress:	We have made significant progress over the past 12 months by: 1) conducting our pre- and post-flight tests, 2) consenting more ISS crew members, 3) updating Institutional Review Board (IRB) applications, 4) submitting amendments and 5) receiving approvals from the University of California San Diego (UCSD) and NASA Johnson Space Center (JSC) Institutional Review Boards. Importantly for this year, we completed our third and fourth, full set of Pre- and Post-Flight Testing of ISS crew members for IVD (intervetebral disk). Highlights for May 2014 to May 2015: • D Chang presented a poster “Risk of Intervertebral Disc Damage after Prolonged Space Flight.” International Olympic Committee, World Conference on Prevention of Injury & Illness in Sport. Apr 10-12, 2014. Monte-Carlo, Monaco. • J Lotz attended ESA Topical Team Follow-On Workshop “Intervertebral Disc Herniations in Astronauts”, June 2014. • J Sayson was featured on NASA TV and represented the NASA IVD Team on “Back Pain in astronauts during and after spaceflight” during the American Astronautical Society's 3rd Annual International Space Station Research & Development Conference, the Hyatt Regency McCormick Place, Chicago, Illinois. June 17, 2014. • J Sayson was Invited Speaker, “Back Pain Mechanisms on the International Space Station: a preliminary report.” at the American Astronautical Society’s 3rd Annual International Space Station Research and Development Conference, Chicago, Illinois, June 19, 2014. • One year Study Debrief with ISS Commander Kevin Ford on 11 Sept 2014, an excellent exchange of data and information by us and him. • Berg-Johansen et al. presented “Bending Properties of Murine Intervertebral Discs Are Reduced Following Spaceflight” at the annual ASGSR meeting in Pasadena, CA, 14 October 2014. • IVD abstract “Post-Flight Herniation of Intervertebral Discs” was accepted by Roscosmos ISS Conf April 2015. • Amended eIRB was approved by JSC. • Jacquelyn Holt received the 2015 Michael L. Pollock Student Scholarship based upon her abstract on muscle. • Alex Snyder was selected for the American Physiological Society (APS) Excellence in Professional Student (MD or

	DO) Research Travel Awards to attend Experimental Biology (EB) 2015 to present his spinal muscle data for IVD. • Shannon Baird, Science & University Prep Teacher at The Preuss School received a summer fellowship to get exposure to space physiology to train her high school science students. • Britta Berg-Johansen presented her research at the ORS 2015 Annual Meeting in Las Vegas and was featured on Orthopaedic Research Society (ORS) Connect for her podium presentation. • Hargens informed Human Research Program (HRP) managers that IVD team may be able to help crew members more with their post-flight rehab of back pain. Two of our three IVD male astronauts are having significant back pain up to one year post-flight. We suggest that NASA hold a Post-Flight Rehabilitation Workshop next year to discuss possible ways for us to share our results with flight surgeons, ASCRs, and vice versa. The Workshop may include ASCRs, flight surgeons, back-pain rehab experts, HRP management, some of the IVD team, and perhaps crew members themselves.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Abstracts for Journals and Proceedings	Chang DG, Sayson J, Chiang S, Riascos-Castaneda R, Walker K, Lotz J, Hargens AR. "Risk of intervertebral disc damage after prolonged space flight." Presented at the IOC (International Olympic Committee) World Conference on Prevention of Injury & Illness in Sport, Monaco, April 10-12, 2014. Br J Sports Med. 2014 Apr;48(7):578-9. http://dx.doi.org/10.1136/bjsports-2014-093494.51 , Apr-2014
Abstracts for Journals and Proceedings	Chang DG. "Head stabilization and neck pain during prolonged space flight." Presented at the 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):286. See http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 ; accessed 5/29/15. , Mar-2014
Articles in Peer-reviewed Journals	Belavy DL, Adams M, Brisby H, Cagnie B, Danneels L, Fairbank J, Hargens AR, Judex S, Scheuring RA, Sovelius R, Urban J, van Dieën JH, Wilke HJ. "Disc herniations in astronauts: What causes them, and what does it tell us about herniation on earth?" Eur Spine J. 2015 Apr 18. [Epub ahead of print] PubMed PMID: 25893331 , Apr-2015
Articles in Peer-reviewed Journals	Bailey JF, Hargens AR, Cheng KK, Lotz JC. "Effect of microgravity on the biomechanical properties of lumbar and caudal intervertebral discs in mice." J Biomech. 2014 Sep 22;47(12):2983-8. Epub 2014 Jul 14. http://dx.doi.org/10.1016/j.jbiomech.2014.07.005 ; PubMed PMID: 25085756 , Sep-2014
Articles in Peer-reviewed Journals	Shymon S, Hargens AR, Minkoff LA, Chang DG. "Body posture and backpack loading: an upright magnetic resonance imaging study of the adult lumbar spine." Eur Spine J. 2014 Jul;23(7):1407-13. Epub 2014 Mar 12. http://dx.doi.org/10.1007/s00586-014-3247-5 ; PubMed PMID: 24619606 , Jul-2014
Articles in Peer-reviewed Journals	Morgan JL, Heer M, Hargens AR, Macias BR, Hudson EK, Shackelford LC, Zwart SR, Smith SM. "Sex-specific responses of bone metabolism and renal stone risk during bed rest." Physiological Reports 2014 Aug 7;2(8):e12119. http://dx.doi.org/10.14814/phy2.12119 ; PubMed PMID: 25107989; PubMed Central PMCID: PMC4246590 , Aug-2014
Articles in Peer-reviewed Journals	Lee SM, Schneider SM, Feiveson AH, Macias BR, Smith SM, Watenpaugh DE, Hargens AR. "WISE-2005: Countermeasures to prevent muscle deconditioning during bed rest in women." J Appl Physiol (1985). 2014 Mar 15;116(6):654-67. Epub 2014 Jan 23. http://dx.doi.org/10.1152/japplphysiol.00590.2013 ; PubMed PMID: 24458754 , Mar-2014
Awards	Holt J, Macias BR, Healey R, Lee SMC, Schneider SM, Snyder A, Chang DG, Hargens AR. "Michael L. Pollock Student Scholarship to Jacquelyn Holt for: WISE 2005: Aerobic and Resistive Exercises Protect Lumbar Paraspinal Lean Muscle Mass during 60 day Bed Rest in Women. American College of Sports Medicine, San Diego, CA, May 2015." May-2015
Awards	Snyder AJ, Macias BR, Healey R, Holt J, Chang DG, Lotz JC, Hargens AR. "American Physiological Society Excellence in Professional Student Research Travel Award to Alex Snyder for: Lumbar Paraspinal Muscle Atrophy during Long Duration Spaceflight. Experimental Biology, Boston, MA, March 2015." Mar-2015
Awards	Snyder AJ, Macias BR, Healey R, Holt J, Chang DG, Lotz JC, Hargens AR. "American Physiological Society Exercise & Environmental Physiology Section's National Space Biomedical Research Institute's Gravitational Physiology Predoctoral Award for: Lumbar Paraspinal Muscle Atrophy during Long Duration Spaceflight. Experimental Biology, Boston, MA, March 2015. " Mar-2015
Books/Book Chapters	Chang DG, Padilla MA, Hargens AR. "Surface Electromyography: Technical Developments and Clinical Applications in Sports Medicine." in "Adaptation Biology and Medicine, New Challenges." Ed. L.M. Popescu, A.R. Hargens, P.K. Singal. New Delhi, India: Narosa Publishing House, 2014. Vol. 7, p. 333-351. 10th World Congress of the International Society for Adaptive Medicine., Jan-2014
Papers from Meeting Proceedings	Berg-Johansen B, Liebenberg EC, Macias BR, Hargens AR, Lotz JC. "Spaceflight Decreases Bending Strength and Alters Failure Mode in Murine Spinal Segments." Presented at the 61st Annual Meeting of the Orthopaedic Research Society, Las Vegas, NV, March 28-31, 2015. ORS (Orthopaedic Research Society) Annual Meeting 2015, Las Vegas, NV, March 28-31, 2015. Paper No: 0208. , Mar-2015